Problem Set 2
Due: Mar. 8, 2021

1. Section 1.3 Problem 16.

   We are given three coins: one has heads in both faces, the second has tails in both faces, and the third has a head in one face and a tail in the other. We choose a coin at random, toss it, and the result is heads. What is the probability that the opposite face is tails?

2. Section 1.4 Problem 22.

   Each of k jars contains m white and n black balls. A ball is randomly chosen from jar 1 and transferred to jar 2, then a ball is randomly chosen from jar 2 and transferred to jar 3, etc. Finally, a ball is randomly chosen from jar k. Show that the probability that the last ball is white is the same as the probability that the first ball is white, i.e., it is $m/(m+n)$.

3. A and B play a series of games. Each game is independently won by A with probability p and by B with probability $1-p$. They stop when the total number of wins of one of the players is two greater than that of the other player. The player with the greater number of total wins is declared the winner of the series.

   (a). Find the probability that a total of 4 games are played.

   (b). Find the probability that A is the winner of the series.

4. Assume that each child is, independently, equally likely to be either a boy or a girl. If every family has three children, what proportion of all sons are eldest sons?